

Appendix B

Data Sets and Programs

Most of the data sets and R programs featured in the book are available at the following web site:

<http://statistics.stanford.edu/~brad>

Follow the link to Data Sets and Programs.

Data Sets

- **Kidney data** (Figure 1.2): 157×2 matrix `kidneydata` giving ages and kidney scores.
- **Prostate data** (Figure 2.1): 6033×102 matrix `prostate`, columns labeled 1 and 2 for Control and Treatment; also `prostdz`, the 6033 z -values in Figure 2.1 and Figure 4.6.
- **DTI data** (Figure 2.4): $15\,443 \times 4$ matrix `DTIdata`; first three columns give (x,y,z) brain coordinates, with z -values in fourth column.
- **Leukemia data** (Figure 6.1a): 7128×72 matrix `leukdata`, columns labeled 0 and 1 for AML and ALL; also `leukz`, the 7128 z -values shown in Figure 4.6 and Figure 6.1a.
- **Chi-square data** (Figure 6.1b): $321\,010 \times 3$ matrix `chisquaredata`, column 1 = gene, columns 2 and 3 conditions 1 and 2, as in Table 6.1 (which is actually gene 23 except that one of the entries for condition 2 has been changed from 1 to 0); also `chisqz`, z -values (6.2) calculated for the 16 882 of the 18 399 genes having at least three sites.
- **Police data** (Figure 6.1c): 2749-vector `policez` of z -values.
- **HIV data** (Figure 6.1d): 7680×8 matrix `hivdata`, columns labeled 0 and 1 for healthy controls and HIV patients; also `hivz`, vector of 7680 z -values obtained from two-sample t -tests on logged and standardized version of `hivdata`.
- **Cardio data** (Figure 8.1): $20\,426 \times 63$ matrix `cardiodata`, first 44 columns

healthy controls, last 19 cardiovascular patients; the columns are standardized but the matrix is not doubly standardized.

- **P53 data** (Figure 9.1): $10\,100 \times 50$ matrix `p53data`, first 33 columns mutated cell lines, last 17 unmutated; also `p53z`, vector of $10\,100$ z -values.
- **Brain data** (Figure 10.7b): vector of $12\,625$ t -values `braintval`; also (x,y) coordinates for smoothing spline `brainsm`.
- **Michigan data** (Table 11.4): 5217×86 matrix `michigandata`, first 24 columns bad outcomes, last 62 good outcomes.
- **Shakespeare data** (Table 11.6): 10×10 matrix `shakedata`.

Programs

- **locfdr**: Produces estimates of the local false discovery rate $\text{fdr}(z)$ (5.2), both assuming the theoretical null hypothesis as at (5.5) and using an empirical null estimate as in Section 6.2 and Section 6.3. Only a vector z of z -values need be entered, but the help file details a list of possible special adjustments. Particularly valuable is the histogram plot superimposing the estimated null and mixture densities $\hat{f}_0(z)$ and $\hat{f}(z)$, which should always be inspected for goodness of fit.
- **simz**: Generates an $N \times n$ matrix X of correlated z -values $z_{ij} \sim \mathcal{N}(0, 1)$, with root mean square correlation (8.15) approximately equaling a target value α ; see the paragraph on simulating correlated z -values in Section 8.2. Comments at the beginning of `simz` show its use. Non-null versions of X can be obtained by adding appropriate constants to selected entries, e.g., the top 5% of rows in the last $n/2$ columns.
- **alpha**: Inputs an $N \times n$ matrix X and outputs estimate (8.18) of the root mean square correlation α . It is best to first remove potential systematic effects from X . For instance, with the prostate data of Section 2.1, one might subtract gene-wise means separately from the control and cancer groups. (Or, more simply, separately apply `alpha` to the 50 control and 52 cancer columns of X .)
- **Ebay**: Produces the empirical Bayes estimated effect size $\hat{E}\{\delta|z\}$ as in Figure 11.1. All that need be provided is the “training data,” the $N \times n$ predictor matrix X and the n -vector Y of zero/one responses, but the help file lists a range of user modifications, including the `folds` call for cross-validation.